

LCD COMMAND CODES

INTERFACING
LCD TO 8051

LCD Command
Codes

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Code (Hex)	Command to LCD Instruction Register
1	Clear display screen
2	Return home
4	Decrement cursor (shift cursor to left)
6	Increment cursor (shift cursor to right)
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
F	Display on, cursor blinking
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to the left
1C	Shift the entire display to the right
80	Force cursor to beginning to 1st line
C0	Force cursor to beginning to 2nd line
38	2 lines and 5x7 matrix



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Program

Assembly Language

Connect the LCD to your 8051 Kit. Then write and run a program to display your name on line 1 of the LCD (first name followed by last name with a space in between).

Note: If you are not monitoring the busy flag of the LCD, put a few milliseconds delay in your program. See the discussion in Chapter 12.

; P1.0-P1.7 are connected to LCD data pins D0-D7
; P2.0 is connected to RS pin of LCD
; P2.1 is connected to R/W pin of LCD
; P2.2 is connected to E pin of LCD

ORG 0000H

MOV A,#38H

ACALL CMNWRT

ACALL DELAY

MOV A,#0EH

ACALL CMNWRT

ACALL DELAY

MOV A,#01H

ACALL CMNWRT

ACALL DELAY

MOV A,#06H

ACALL CMNWRT

ACALL DELAY

MOV A,#80H

ACALL CMNWRT

ACALL DELAY

MOV A,#'V'

ACALL DATWRT

ACALL DELAY

MOV A,#'I'

ACALL DATWRT

ACALL DELAY

MOV A,#'T'

ACALL DATWRT

ACALL DELAY

AGAIN: SJMP AGAIN

CMNWRT:

MOV P1,A

CLR P2.0

CLR P2.1

SETB P2.2

CLR P2.2

RET

DATWRT:

MOV P1,A

SETB P2.0

CLR P2.1

SETB P2.2

CLR P2.2

RET

DELAY: MOV R3,#0FFH

MOV R4,#0FFH

MOV R5,#55H

HERE: DJNZ R5,HERE

HERE1: DJNZ R4, HERE1

HERE2: DJNZ R3,HERE2

RET

END